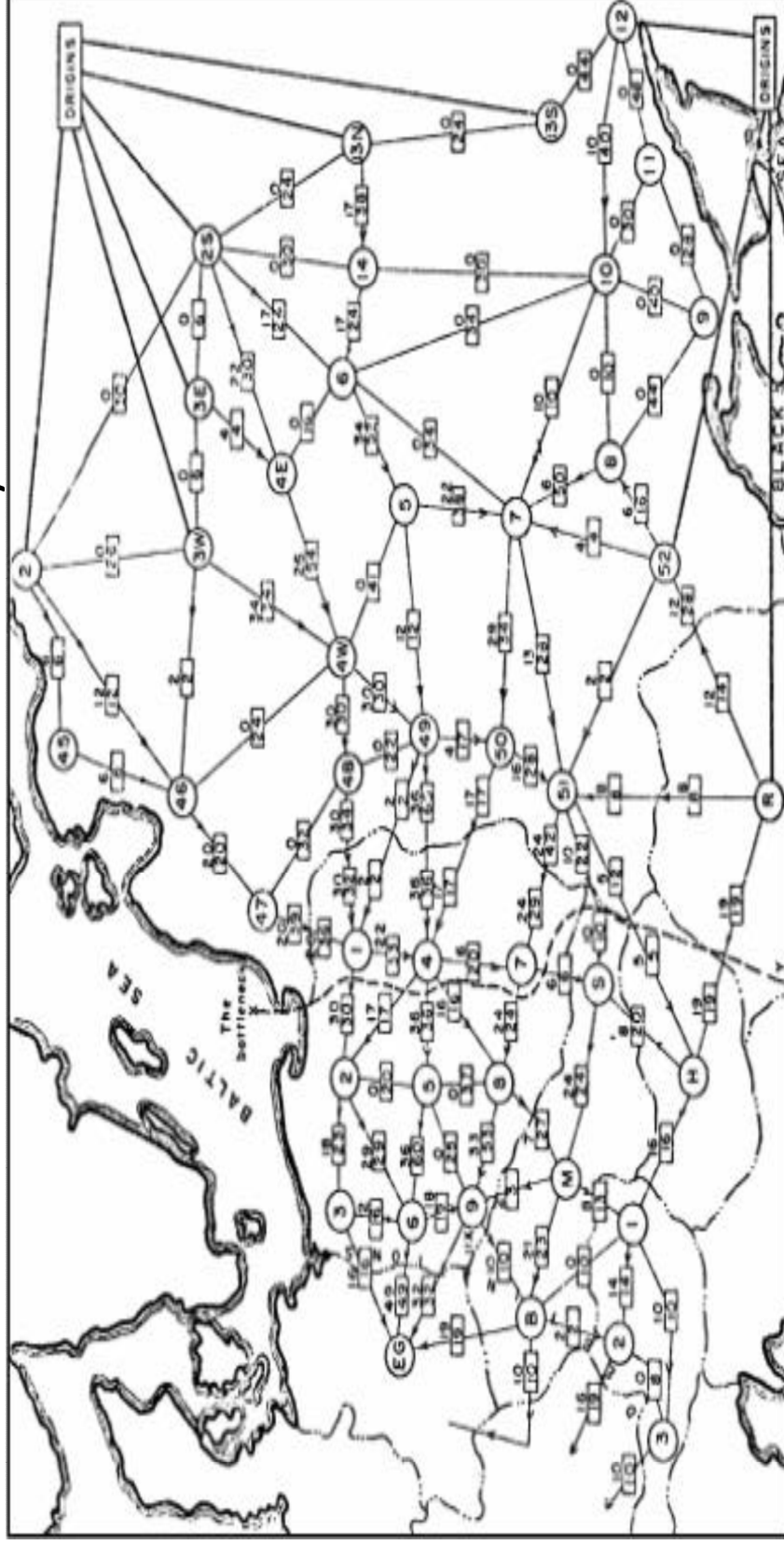


Brief Context

(Why do we care?)

- **Question:** What is the optimal way to disrupt the rail network between the Soviet Union and Eastern Germany?

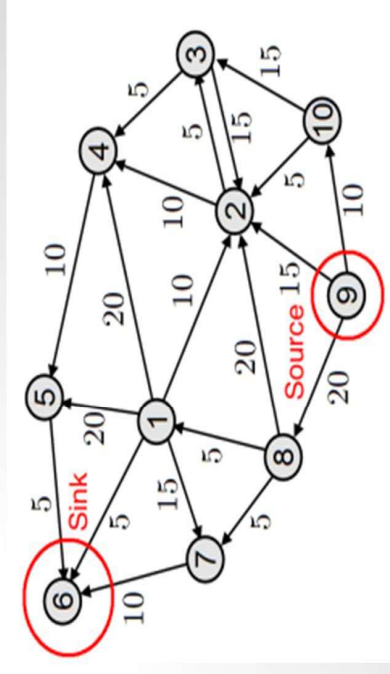
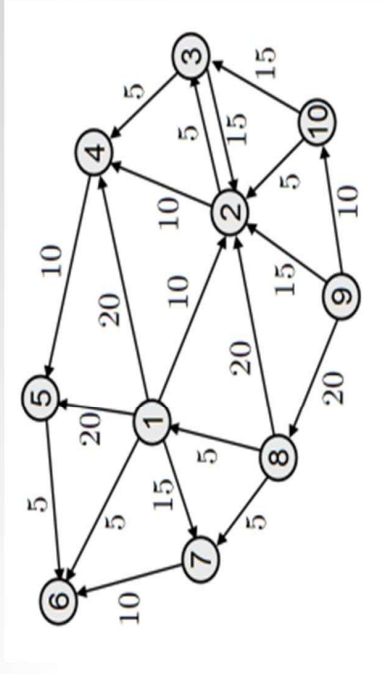


Agenda

1. Basic Background
2. Cuts
3. Flows
4. The Max-Flow, Min-Cut Theorem
5. Conclusion

Basic Background

- A directed, weighted graph is called a (flow) **network**.
- Each edge has a weight and direction.
- We assume there exists a **source** and a **sink**.



A **network** N is a set containing:

- a directed graph $G(V, E)$;
- a vertex $s \in V$ which has only outgoing edges, we call s the source node;
- a vertex $t \in V$ which has only incoming edges, we call t the sink node;
- a positive capacity function $c : E \mapsto \mathbb{R}^+$.

A **flow** f on a network N is a function $f : E \mapsto \mathbb{R}^+$. Flow f is a **feasible flow** if it satisfies the following two conditions:

1. **Edge capacity limit:**

$$\forall e \in E, 0 \leq f(e) \leq c(e)$$

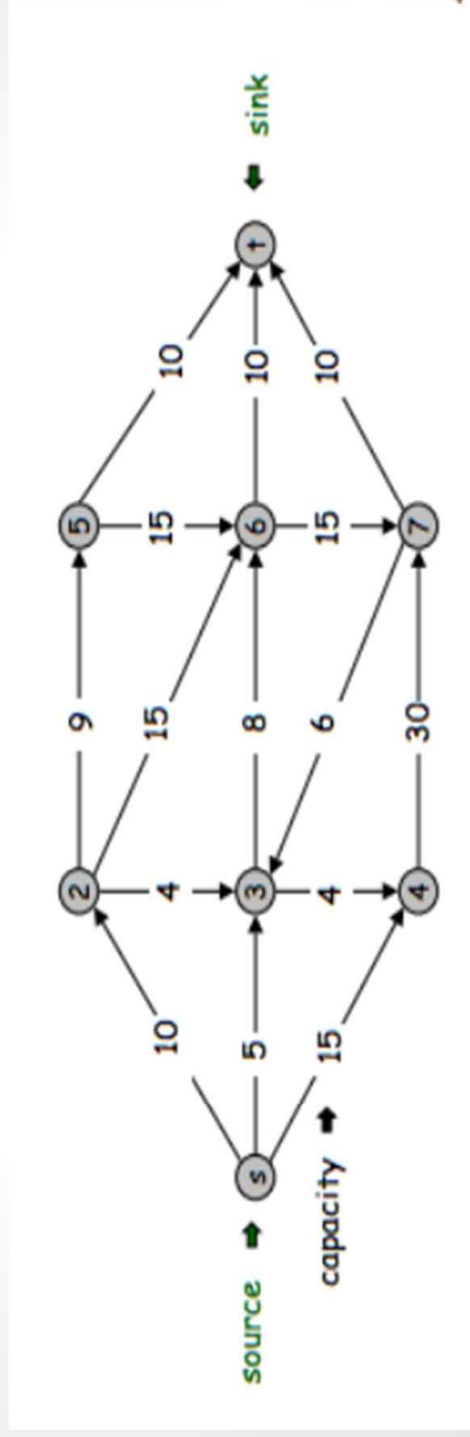
2. **Conservation of flow:**

$$\forall v \in V \setminus \{s, t\}, \sum_{e \text{ leaving } v} f(e) = \sum_{e \text{ entering } v} f(e).$$

The **value** of a flow f is defined as

$$v(f) \equiv \sum_{e \text{ leaving } s} f(e).$$

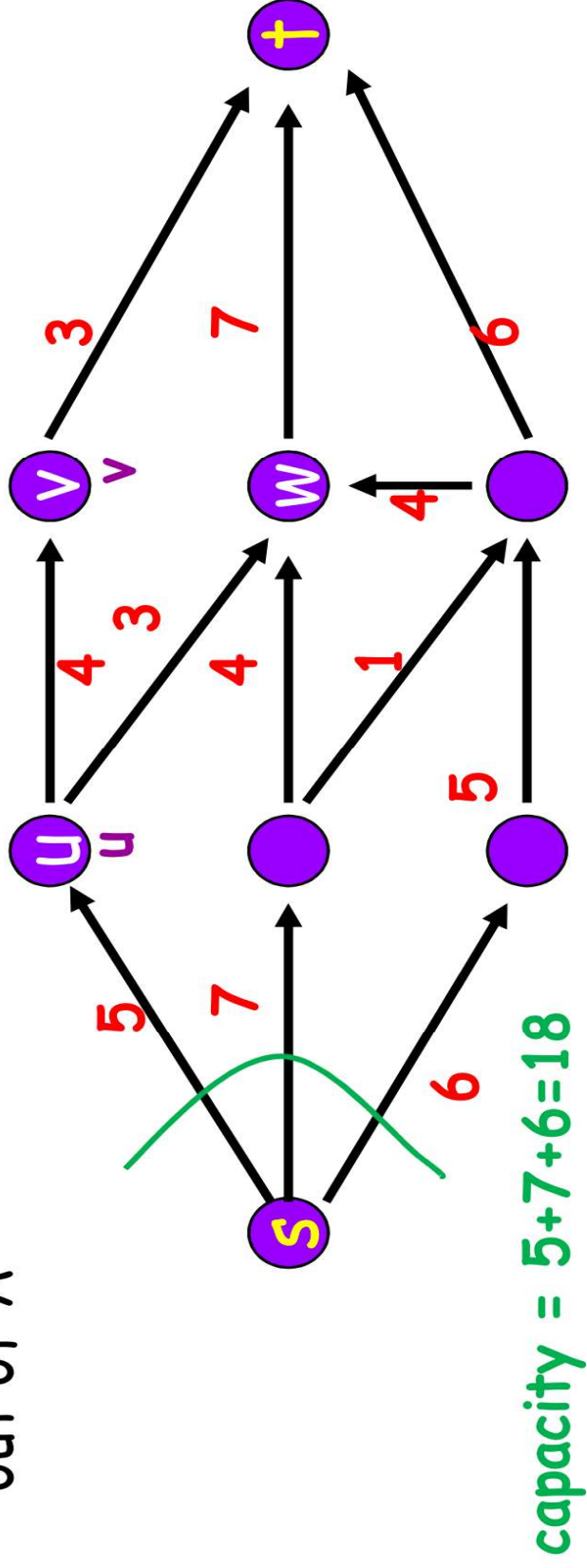
Consider a basic flow network.



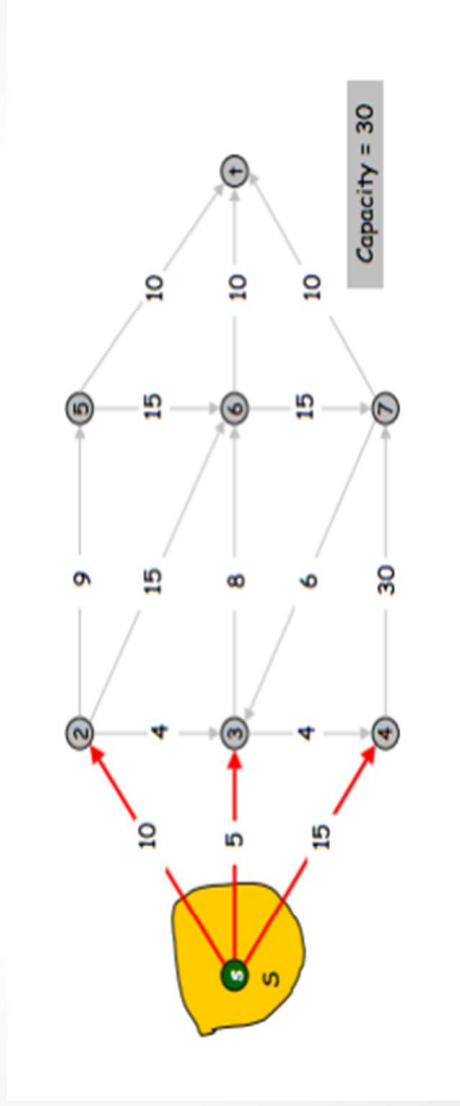
Cuts

- A **cut** is a partition of the vertices into disjoint subsets S and T . In a flow network, the source is located in S , and the sink is located in T .
- The **cut-set** of a cut is the set of edges that begin in S and end in T .
- The **capacity** of a cut is sum of the weights of the edges beginning in S and ending in T .

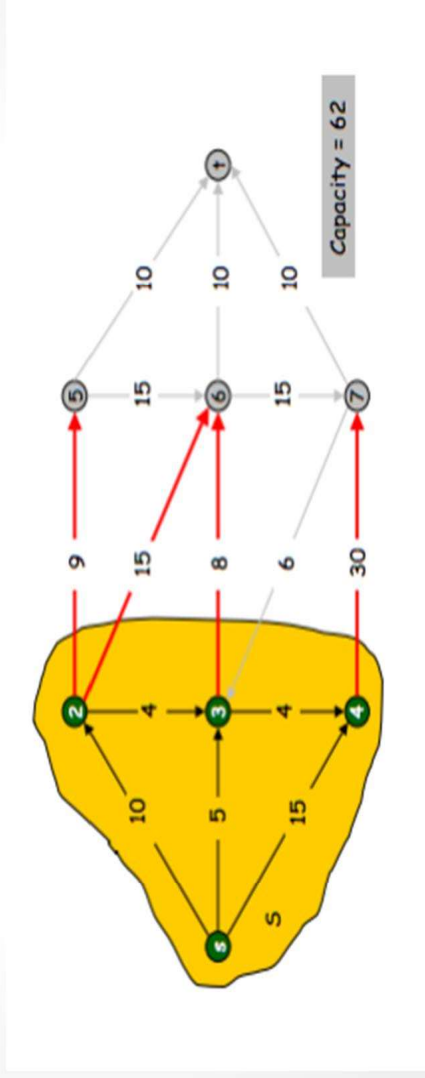
- an s - t cut is a partitioning of V in two blocks, A and B , with $s \in A$ and $t \in B$
- the capacity of an s - t cut is the sum of edge capacities out of A



Cut Examples (cont'd)



Cut Examples (cont'd)

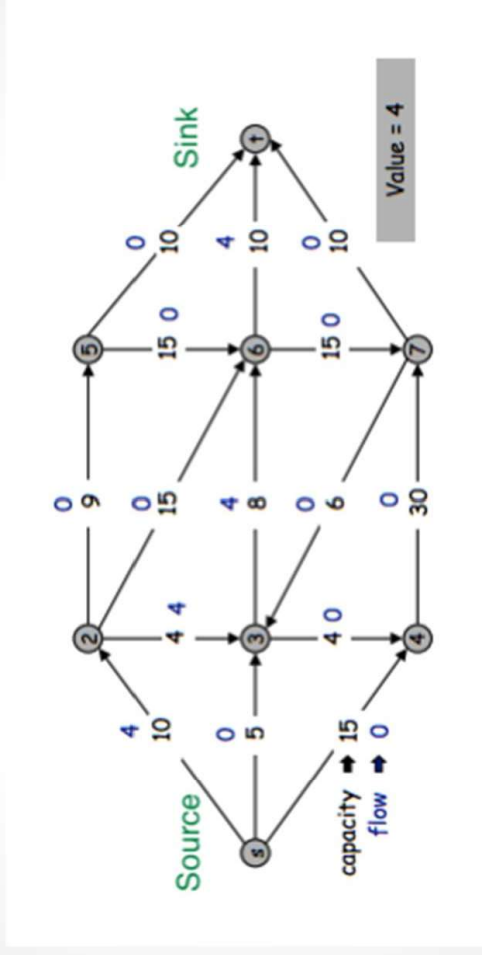


Flow

The **flow** over a network is a function $f: E \rightarrow \mathbb{R}$, assigning values to each of the edges in the network which are nonnegative and less than the capacity of that edge. For each intermediate vertex, the outflow and inflow must be equal.

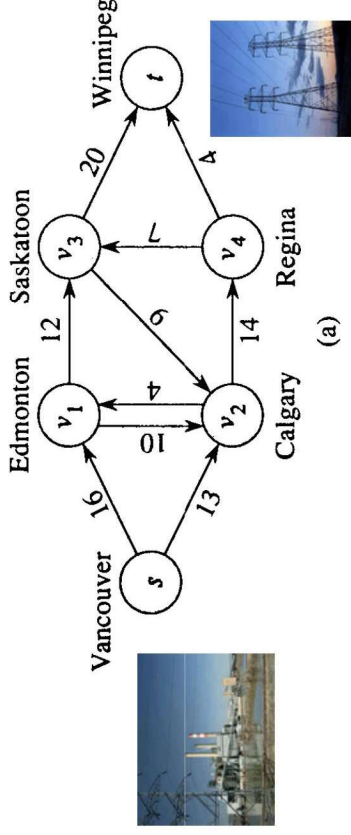
The value of this flow is the total amount leaving the source (and thus entering the sink).

Flow Example



an example of a flow network

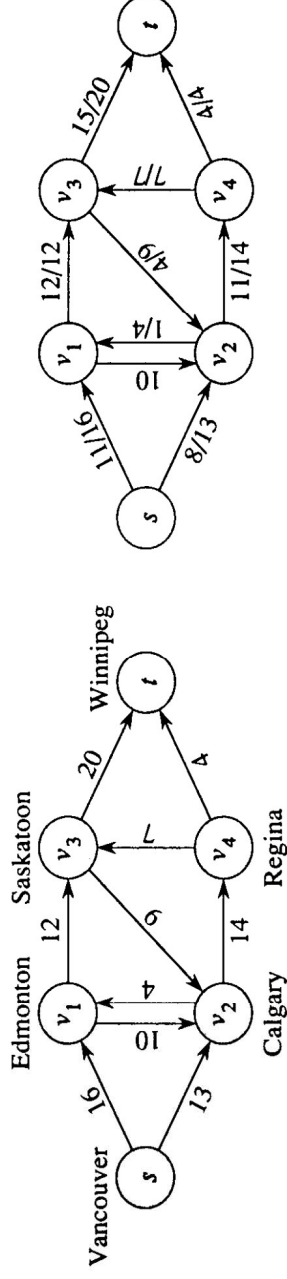
- suppose we operate a electrical power distribution company in Canada.
- the electrical power is generated in Vancouver, and as much as possible power is to be transported to Winnipeg:



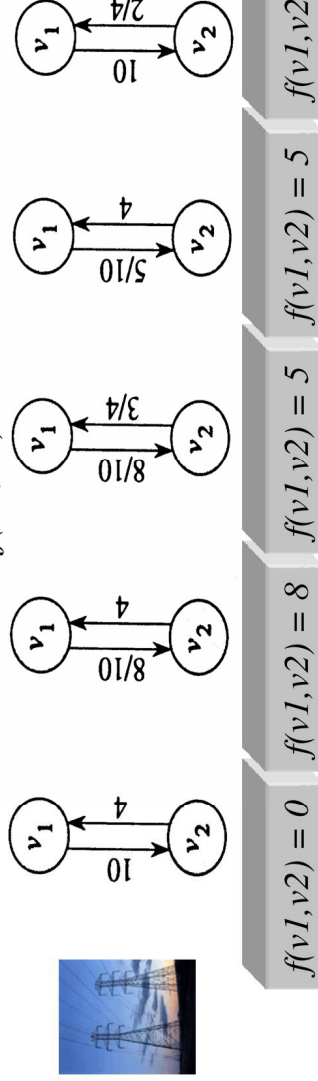
- high-voltage cables connect the cities, each of the cables has a maximum transport capacity in megawatt
- note that between Edmonton and Calgary two cables exist:
 - $c(v1, v2) = 10$ that can transport 10 megawatts from Edmonton to Calgary
 - $c(v2, v1) = 4$ with a capacity of 4 megawatt in the opposite direction.

a possible flow assignment

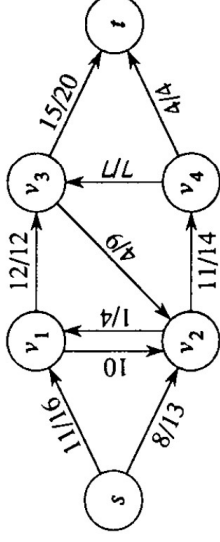
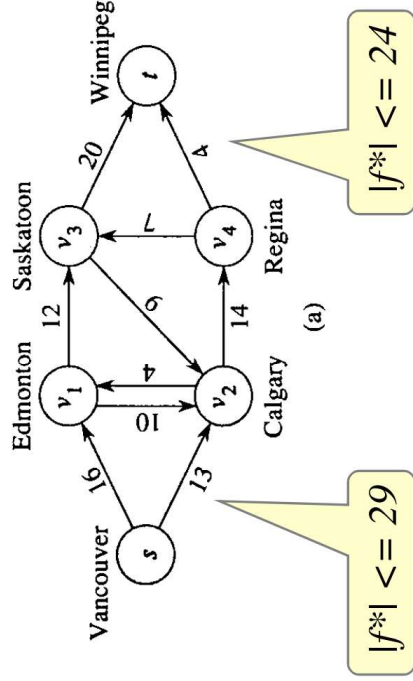
- the figure on the right shows a possible flow f where $|f| = 19$, meaning 19 megawatts are transported from Vancouver to Winnipeg:



- if there is more than one edge between two vertices, certain flows in the edges can cancel out: we care about the **net flow** $f(v_1, v_2)$ between vertices:



now find the maximum flow $|f^*|$



A possible flow with $|f| = 19$

- if we look at the edges connected to the source and sink:
 - the flow out of Vancouver is never $> 16 + 13 = 29$
 - the flow into Winnipeg is never $> 20 + 4 = 24$
- so... is the maximum flow through the network then 24 MW???

Flow Example (cont'd)

